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An Investigation of the Remanence Characteristics of Three Sedimentary Units of the Silurian Mascarene Group of New Brunswick, Canada

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Thermal, chemical, alternating field (AF) and two-stage (AF followed by thermal) experiments were performed on 18 sites (217 specimens) of three sedimentary units (groups A, B and C) of the Mascarene Group of southwestern New Brunswick. Groups A and B of Pembroke age (late Late Silurian) apparently carry a dual polarity initial remanence yielding a pole at 087°E, 05°S. Group C of probably Waweig age (older than Pembroke) carries a magnetite remanence and a hematite remanence with the former probably being the initial and yielding a pole at 085°E, 28°N. This would indicate a pole shift of some 30° southwards relative to the sampling area during the Upper Silurian, in agreement with the hypothesis that considerable polar shift took place during the Siluro-Devonian period. The extensive experimental work (over 4000 treatments) demonstrates the need to use several techniques in the pursuit of the magnetic history of the rock. For example, if the work had been limited to thermal treatment up to 650°C, an overprinted or more likely an apparent remanence could conceivably have been mistaken for the initial remanence. Conversely, the considerable information gathered through the use of different techniques allows one to advance and model the most probable scenario of remanence acquisition while contributing additional knowledge about the magnetic characteristics of sedimentary units. Results from this investigation, including those of a Mascarene Group intrusion also reported here, are discussed in the context of Siluro-Devonian poles of North America. Different options are open to the interpreter attesting to the difficulties in establishing both the rock and remanence ages. It is concluded that more data from detailed investigations are needed before the evolution of the eastern seaboard and the Appalachians can be retraced by paleomagnetic means.

INTRODUCTION

Paleomagnetic results from North America for the Cambrian to Devonian periods are few. Prior to 1977, only 20, 5, 3 and 6 poles from Cambrian, Ordovician, Silurian and Devonian rock units, respectively, could be found in the literature. Many of those poles were obtained from natural remanent magnetization (NRM) or after limited experimental work and their reliability is questionable. More recently (1977–1979), there has been an influx of new paleomagnetic studies (20–25 Cambrian to Devonian poles), references to most of which can be found in the work of *Roy et al.* [1979]. To be added are *French and Van Der Voo* [1977], *Brown* [1979], *Gillett and Van Alstine* [1979], *Kent* [1979], *Van der Voo et al.* [1979] and *Watts and Van der Voo* [1979].

With the development of new technical and analytical methods, the thoroughness of the new investigations has generally improved and has shown that multi-component remanences are common in Paleozoic rocks [*Van der Voo et al.*, 1976]. This is particularly true for rocks which may have been exposed to considerable tectonic activity, e.g., Appalachian rock units.

As more data points are added to the scanty data base, it is becoming evident that the Lower Paleozoic picture is much more complex than previously believed. Is the complexity attributable to a fragmented North American [*Morris*, 1976; *Kent and Opdyke*, 1978] or to an intricate polar wander path relative to the whole of North America? Many more data from different parts of the continent are needed to confirm the integrity or nonintegrity of North America. If the latter is correct, then contemporaneous poles from each fragment will be required to determine the timing and the extent of relative motion.

This investigation contributes more data points from the Atlantic seaboard and also emphasizes that, without the benefit of extensive experimental and analytical work, certain remanences could easily remain unnoticed, thereby providing one with a simpler but possibly erroneous interpretation. This is most visible in the group A and B results. Two directions (*P* and *R*) are clearly identified as shown first in the presentation of the experimental data (section: 'Groups A and B') and the following section ('Compilation of Results'). Then it is shown in subsequent sections ('The Magnetic History and a Hypothetical Model') that all data can be satisfied only by assuming that *P* represents the initial remanence while the more readily available *R*, which could have been mistaken for the initial remanence had the experimental work been more limited, is most probably an apparent remanence; that is, it does not even represent any real field direction but is the sum of two or more remanences. The implication is serious since *R* would have yielded a pole in rough agreement with other Siluro-Devonian poles. It is noted that the interpretation can be advanced only owing to the considerable amount of varied information made available through the use of detailed thermal, chemical and alternating field (AF) treatments.

GEOLOGY AND SAMPLING

Lower Paleozoic rocks underlie the Passamaquoddy Bay region of southwestern New Brunswick (Figure 1). Although the oldest rocks of the region are the Lower Ordovician Cookson Formation of the Charlotte Group, most of the bedrock is formed of Silurian and Devonian rocks. The Silurian succession is represented in New Brunswick by the Mascarene Group which comprises several formations (in ascending order): Oak Bay, Quoddy, Waweig, Pembroke and Eastport [*Cumming*, 1967; *Pickerill*, 1976]. Approximate age limits of 420 to 400 Ma are given by *Pajari et al.* [1974]. The uppermost Eastport Formation may however be lowermost Devonian (Gedinnian ≈ 392 Ma) in age [*Pickerill and Pajari*, 1976]. The

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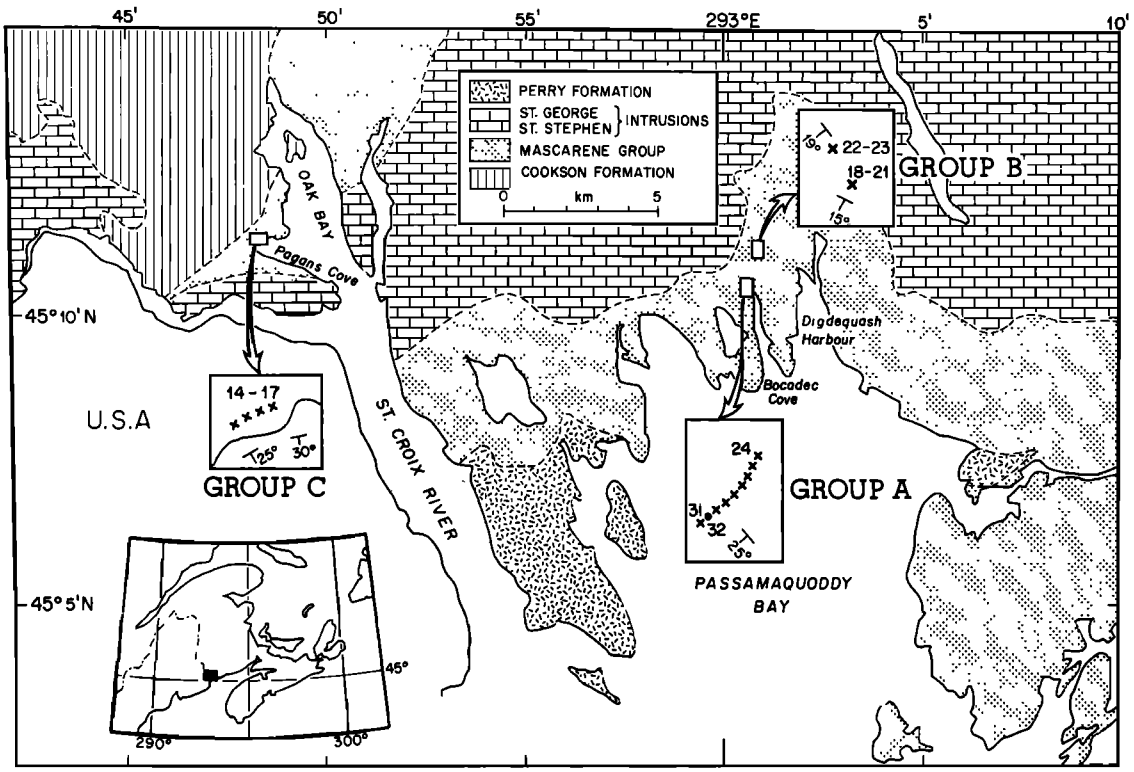


Fig. 1. Generalized map of southwestern New Brunswick, Canada, with localities of three groups (A, B and C) of sampled sites with bedding attitudes.

Devonian rocks are represented by the Lower Devonian St. Stephen and St. George plutons and the Upper Devonian Perry Formation. Both K-Ar and Rb-Sr age determinations yield a lowermost Devonian age (≈ 390 Ma) for the St. George pluton [*Wanless et al.*, 1973; *Pajari et al.*, 1974]; the St. Stephen pluton is believed to be slightly older. *Pajari et al.* [1974] have suggested that the last phase of the Acadian deformation in the area occurred between 394 and 410 Ma and regard the Mascarene Group as a pre-Acadian unit and both the St. Stephen and St. George plutons as post-Acadian intrusions. Paleomagnetic results have been obtained from the Perry Formation [*Black*, 1964; *Robertson et al.*, 1968], the St. George pluton, the St. Stephen pluton and 3 intrusions of the Mascarene Group [*Roy et al.*, 1979].

The Mascarene Group consists of acid and basic volcanics and associated clastic sedimentary rocks: conglomerates, limestones, red shales, brown to grey siltstones, grey and red sandstones, argillites, arkoses and graywackes. The area represented in Figure 1 has been the subject of much detailed mapping which has demonstrated that the geological setting is complex [*Cumming*, 1967; *Hay*, 1967; *Ruitenberg*, 1967, 1968, 1972]. The exact age relationship among the different intrusions and sedimentary units is not easily established, partly because of folding during the Acadian Orogeny. In some areas, *Ruitenberg* [1967] has recognized four phases of deformation. It is not in the scope of this paper to describe in detail the geology of the area and readers are referred to the detailed work of the authors referenced in this paragraph. It is however clear that valuable additions to the tectonic history could be obtained from an extensive and exhaustive paleomagnetic survey of the Mascarene-Nerepis Belt. For the time being, the sampling was limited to three specific areas (groups A, B and C) as shown in Figure 1. Groups A and B consist of stratigraphic successions of 8 and 6 sites, respectively, while group C consists of 4 sites within 2 meters stratigraphically. At group A sites, the beds are dipping 25° at 230°; at group B sites, they are dipping 19° at 230° and 15° at 208°. At group C sites, the dip of the beds varies from 25° to 30° with azimuths varying from 150° to 170°.

Group A and B rocks are siltstones of unit 5 of *Ruitenberg* [1968], unit 6 of *Hay* [1967] and unit 9 of *Cumming* [1967]. Each author correlates his unit with the Pembroke Formation. Group C rocks are feldspathic graywackes of unit 11 of *Cumming* [1967] and unit 4c of *Ruitenberg* [1968]. They belong to a belt of undifferentiated Silurian strata in the Oak Bay area which have been tentatively correlated with the Dennys and Edmunds Formation (below the Pembroke Formation) of Maine [*Cumming*, 1967] which in turn are probably the equivalents of the Waweig Formation of the Piskahegan–Rolling Dam area of New Brunswick [*Ruitenberg*, 1968]. Therefore, although the exact age of these strata has not been ascertained, it would appear that they are older than the group A and B stratigraphic sequences. According to *Ruitenberg* [1968], the rocks in the section to the west and northeast of Oak Bay have been slightly metamorphosed with the grade increasing gradually towards the granite contact to the south (Figure 1). 'The alteration is chiefly due to increased biotite content of the more pelitic beds, which are separated by partly recrystallized more siliceous bands' [*Ruitenberg*, 1967, p. 87]. He suggests that the metamorphism occurred at about the same time as the second phase of deformation—Early to Middle Devonian. As discussed later, the ages of both the de-

formation of the strata and the remanence acquisition are most important for the interpretation of the group C paleomagnetic data.

Samples were also taken from a badly fractured intrusion (site 32) cutting group A sediments. The intrusion belongs to unit 7 of *Hay* [1967] and unit 6 of *Ruitenberg* [1968]. The age of those units is uncertain, but is probably younger than the Pembroke Formation.

METHODOLOGY

From each of the 19 sites, 3 to 8 cores (2.5 cm diameter) were drilled in the field to yield a total of 140 cores, with 1 or 2 specimens (height of 2.2 cm) cut from each core. Experiments were carried out on 225 specimens in quasi-zero residual fields (<2 nT except for chemical experiments <100 nT). Three basic treatments were used: thermal, chemical and AF. Combinations of treatments (e.g., AF followed by thermal or vice versa) were also frequently used to separate the different remanences. Each specimen was subjected to detailed experimental work. Specimens from the same core were usually subjected to different treatments often yielding different results and, for this reason, specimens (not cores) are given unit weight in the within-site statistical analysis. Specimens were treated in batches of 25. This is referred to as a 'run.' The schedule for the initial run is described below for each treatment. In subsequent runs, the schedule was often changed in order to obtain more information; e.g., the 625°C step was replaced by 610°, 620°, 630° and 640°C steps.

For the initial run, these were the schedules. For the thermal experiments, incremental steps of 100°C were used up to 400°C and thereafter 450°, 500°, 550°, 600°, 625°, 650°, 660°, 680° and 690°C. For the chemical experiments, the specimens were immersed 15 times in hydrochloric acid (HCl) for increasing periods of time until a cumulative time of 4500 hours was reached. Acid concentration of 8 *N* was used for the first 700 hours and 10 *N* thereafter. The experiments were performed at a pressure of 10 kPa. For the AF experiments, incremental steps of 10 mT were used up to 150 mT and thereafter in steps of 25 mT up to 275 mT, and 290 mT (1 millitesla = 10 oersteds). However, except for the initial run, most of the specimens after the 100–140 mT treatment were transferred to the thermal treatment schedule described above. Each specimen was subjected to 12–30 treatments so that over 4000 treatments and measurements were performed in the course of the investigation. Treatments were discontinued only when it became obvious after 3 or 4 treatments that no significant amount of ancient remanence of any stability remained in the specimen or alternately the remaining remanence was marked by large parasitic remanences acquired by various means from treatment to measurement as described below.

Two types of instruments were used for the measurements: a reprogrammed digital spinner magnetometer (DSM) and an astatic magnetometer with negative feedback [*Roy et al.*, 1972]. As shown in the vector diagrams, the last point is not at the origin but is considered as the last significant vector for the determination of an ancient remanence. Further readings were obtained after subsequent treatments (e.g., 700°C) since an instrument does not measure zero remanence if only because of the noise level although, in many instances (group C especially), the measured intensity was above the noise level of the instrument. Three rejection criteria were used for estimating the unreliability of the measured vector as an indicator of an ancient field:

1. One criterion is measurements with intensities lower than 0.3 mA m^{-1} (1 mA m^{-1} = 10^{-6} emu). These were regarded as insufficiently accurate and beyond the resolution of the measuring instruments.
2. The second criterion is the occurrence of large irregular vector changes in both direction and intensity after successive treatments with small incremental steps (e.g., 5–10 mT). These were attributed to parasitic remanences acquired during AF treatment or during exposure to the laboratory ambient field between treatment (thermal or AF) and measurement.
3. The last criterion is the occurrence of large temporary induced magnetizations acquired during measurement. These can be detected by two complementary observations: the statistical parameter used to assess the degree of uniformity (or nonuniformity) of within-specimen direction of magnetization [*Roy and Robertson*, 1978; *Paveck and Lapointe*, 1978] and the behavior of the instrument. With both (astatic and spinner) magnetometers, the occurrence of such magnetizations produces a parameter value indicating a high degree of nonuniformity. In addition, the astatic magnetometer reacts as if in an unstable mode while the spinner magnetometer requires a much longer (automatically selected) spin time than usually needed for specimens of comparable intensities.

RESULTS

The results of groups A and B are presented together. No clear distinction can be made about the magnetic behavior at the group level. In general, the specimen magnetic carriers react equally differently according to the treatment applied; however, the magnetic response to a particular treatment is remarkably similar for specimens of both groups. The description and observations (with respect to bedding) of the magnetic behavior discussed below and illustrated in the figures are therefore applicable to specimens and sites of both groups with a few exceptions as noted. Some of the figures fulfil two functions: (1) presentation of the observations (closed symbols) and (2) the results of vector subtraction (open symbols). Therefore, in the initial presentation of the observed results ('Groups A and B' section), the open symbols should be ignored. Results of vector analysis are discussed in subsequent sections but only to demonstrate the probable existence of a remanence of opposite polarity. For reasons given later, they have not been used to determine the direction of that remanence.

The results of group C are quite different and are presented in a later section following the presentation of results obtained from the intrusion cutting the group A sedimentary unit.

To facilitate identification of the treatments in the vector diagrams, all thermal, chemical and AF results are represented by dots, triangles and squares, respectively.

GROUPS A AND B

Two directions were revealed during the different treatments: a positive steeply inclined direction (*P*) and a shallowly inclined direction (*R*).

The NRM intensities are fairly constant with over 90% being between 5 and 20 mA m^{-1}. The site mean directions are steeply inclined (about 192°, +68°) and in good agreement except for site 24 where the directions are scattered (Table 1). Within sites, the directions are reasonably well grouped and significantly different from the direction of the present earth's field (340°, +75°) indicating that the remanence is composed

TABLE 1. Analysis of Directions Before (NRM) and After (*P*) Comprehensive Thermal, Chemical and AF Treatments

		NRM					High Stability Remanence (*P*)				
Site	SH	N_1	D°, I°	κ	α_{95}	Pole Long., Lat.	N_2	D°, I°	κ	α_{95}	Pole Long., Lat.
						Group A					
31		11	254, +66	86	06	069°E, 23°S	11	240, +62	15	12	073°E, 13°S
30	8	11	269, +74	19	11	074°E, 37°S	11	247, +61	14	12	068°E, 16°S
29	4	17	207, +62	12	11	094°E, 02°S	15	240, +61	15	11	072°E, 13°S
28	16	14	157, +61	03	30	090°E, 01°N	10	220, +68	07	20	088°E, 12°S
27	1	16	224, +51	21	08	076°E, 04°N	10	222, +44	24	10	076°E, 10°N
26	18	22	157, +61	14	09	130°E, 00°S	15	208, +57	09	14	091°E, 04°N
25	11	11	123, +57	05	23	155°E, 07°S	11	182, +73	09	17	112°E, 13°S
24	3	9*	229, +02	01	76	... ...	9	230, +39	08	20	067°E, 09°N
Mean		7	192, +70	12	18	106°E, 10°S	8	226, +59	30	10	079°E, 05°S (12°, 15°)†
						Group B					
18	(300)	17	192, +67	139	03	106°E, 05°S	12	188, +69	16	11	108°E, 08°S
19	3	16	163, +68	145	03	124°E, 08°S	16	190, +67	20	8	107°E, 05°S
20	2	12	181, +56	50	6	112°E, 08°N	10	183, +60	22	10	111°E, 03°N
21	1	13	186, +54	50	6	108°E, 10°N	2*	196, +72	...	...	104°E, 13°S
22	60	16	250, +80	05	17	090°E, 36°S	12	225, +67	16	11	085°E, 13°S
23	6	9	216, +61	17	13	087°E, 03°S	8	206, +60	08	20	094°E, 01°N
Mean		6	192, +66	34	12	105°E, 04°S	5	198, +65	94	08	101°E, 04°S (10°, 13°)†
Average		13	192, +68	19	10	105°E, 07°S	13	216, +62	33	07	087°E, 05°N (08°, 11°)†

SH indicates the relative stratigraphic heights in meters with site 31 as the youngest; the distance between the two sections would be ≃300 m assuming continuity. D° and I° are the east declination and the inclination, respectively. κ is the precision estimate. α_{95} is the radius of circle of confidence at probability level of 0.05. N_1 is the number of specimens treated. N_2 is the number of specimens yielding a steeply inclined remanence after treatment of 4500 hours in acid, 630°–670°C, 150–290 mT or a combination of the latter two. All directions given with respect to bedding (corrected for tilt). Without correction for tilt, the NRM means are 104°, +75° (pole 146°E, 33°S) for group A and 158°, +79° (pole 122°E, 23°S) for group B, both significantly different from the present earth's field direction.

* Not used for 'mean' because of large α_{95} or small number of determinations.

†The semi-axes (δp, δm) of the oval of confidence at probability level of 0.05.

principally of one or more ancient remanences with very little if any having been produced in recent times.

Thermal treatment. The magnetic vectors initially move to lower inclinations (*R* direction) (most of them ranging from +40° to −20°) accompanied by a large decrease in intensity (Figures 2 and 3). The temperatures at which the low inclinations occur vary slightly from site to site but, in general, are most prominent in the 400° to 630°C range. Upon further treatment, the vector steepens (positively) with an increase in intensity and then stabilizes in direction in the 660° to 690°C range (*P* direction); the intensity reaches a maximum (≃675°C) and afterwards decreases until, at 700°C, all stable remanence either disappears (criterion 1; Figure 3) or is masked by induced magnetizations (criterion 3; Figure 2 caption). This pattern is followed by almost all specimens (sites 21, 30 and 31 excluded) with only slight variations in the temperature at which the low inclinations (*R*), the steep inclinations (*P*) and the last significant point occur. However, the low inclination hiatuses are marked by highly variable declinations among sites (Table 2; 116°–172° for group A and 155°–188° for group B) and, to a lesser extent, even within site.

At sites 30 and 31, there are no indications of low inclination vectors (*R*). The NRM vectors are steeply inclined and upon treatment only small changes occur in direction while the intensities smoothly decrease towards zero, the unblocking temperature (T_{UB}) [*Roy et al.*, 1979] spectrum being distributed mostly in the 100°–675°C range. Apparently, a single remanence (*P*) is carried at these two sites. This conclusion is supported by the chemical and AF experiments which yield similar results. At site 21, the vectors generally move to an intermediate inclination (+38°) where they remain from 400° to 630°C. This is followed by a slight increase (5°–15°) in inclination until the remanence disappears at about 650°C.

Chemical treatment. The results (1 to 4 specimens/site) of this treatment are very consistent. They reveal a single *P* remanence without any tendency towards the *R* direction. Most specimens have steeply inclined NRM's, very little change in direction occurs, and the intensity gradually decreases (Figure 4). For specimens with shallower NRM inclinations, a steepening of the inclination takes place during the first 700 hours of leaching, and, thereafter, the vector follows the pattern described by the other specimens.

At least one specimen was sliced after each interval in acid to verify acid penetration. During the first 950–1150 hours, a charcoal gray (to purple) colored cement(?) is removed to uncover a buff colour which upon further leaching gradually changes to a paler and paler pink color. The contrast between the charcoal gray and buff colors is fairly sharp and easily identifiable while the change from buff to pink is subtle. The disappearance of the charcoal grey cement corresponds to a reduction of about 60% of the NRM. However, it is improbable that the cement accounts for as high a percentage of the total remanence. These rocks are fine grained and well compacted so that the penetration of acid is not rapid enough to ensure total destruction of the cement without at least partially attacking other magnetic carriers especially those in the outer zone of the specimen. This means that, in this instance, the treatment is not totally effective. Assuming that the carriers of different phases of remanence acquired from deposition to lithification can be successively destroyed [*Roy and*

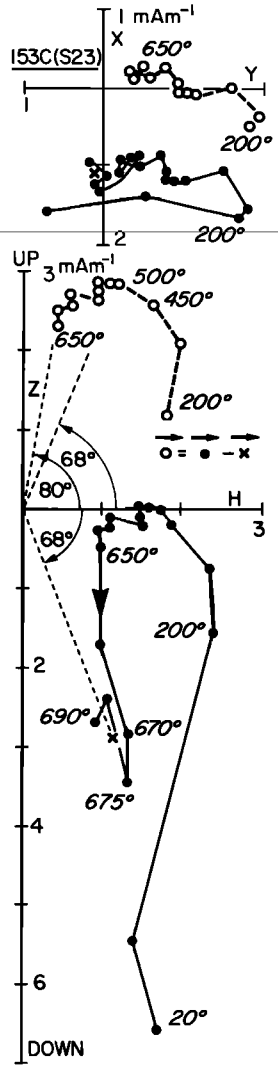


Fig. 2. Vector diagrams showing changes in both direction and intensity after treatment at successively higher temperatures. The solid dots represent the termination of the observed magnetic vector. The horizontal component of the magnetic vector is shown in the upper diagram; the declination D is given by the angular distance from the positive X axis and the intensity is given by the distance of the dot from the origin. The magnetic vector of the total intensity is represented in the lower diagram as a function of the horizontal (H) and vertical (Z) components; the total intensity is given by the distance of the dot from the origin and the inclination ($\pm I$) by the angular distance from the H axis. The open circles show the (sought) vectors obtained by subtracting the mean of the 670° to 690° vectors (represented by X) from each of the 200° to 650°C observed vectors. This combination of vector diagrams is particularly useful for fully illustrating the results of vector subtraction. Hence it is shown that the 500° to 650°C sought vectors all have an inclination (−68° to −80°) very nearly opposite to the inclination (+68) of the high unblocking temperature (T_{UB}) remanence represented by the 670° to 690°C vectors. The horizontal component is not opposite to the subtracted vector but biased to the east (+Y axis). However, its intensity is much smaller (3 to 8 times at 500° and 650°C, respectively) than the vertical (Z) component so that the departure from antiparallelism of the two vectors is small. The significance of this bias is discussed later in the text (and Figure 8). (Note: in this and subsequent figures, for purposes of figure clarity, not all dots are identified. For the same reason, some intermediate results have not been plotted. For example, for this specimen only 9 of the 13 vectors observed between 500° and 650°C are shown. The dots do not terminate at the origin. Although further treatments were performed, the last dot is deemed to represent the last significant observation according to the criteria described in the text. The number identifies the specimen (and the site.)

Park, 1974], the low porosity to acid may cause it to happen in an overlapping sequence throughout the specimen. Therefore it is possible for a phase—even though its initial intensity may be larger than the individual intensities of other phases—to remain unobtrusive throughout the treatment. This probably accounts for the apparent absence of the R direction using this treatment as discussed later.

AF treatment. The magnetic vector usually moves from a steep to a low inclination (R) in the 50 to 100 mT range. Upon further treatment, the vector generally becomes negatively inclined, sometimes with an intermediate to steep inclination (Figure 5), sometimes with a shallow inclination (Figure 6). Only in a few instances does the vector return to a steep positive inclination, the pattern described by the thermally treated specimens.

For the specimens treated up to 290 mT, the vector frequently moves to negative inclinations, but, in the 150–290 mT range, large irregular changes (up to 20°) occur from treatment to treatment. The directions keep hovering in intermediate inclinations (−30° to −60°) and declinations between 060° and 120° as if the vector is attempting to reach the negative polarity of the P direction. However, this is apparently prevented by the acquisition of parasitic remanences (criterion 2) and/or the ineffectiveness of the treatment to separate the negative from the positive polarity remanences owing to overlapping of coercivity spectra. The former cannot be discounted since in this range the intensities (0.5–1.0 mA m^{-1}) are nearing the limit of the instrument and account for less than 10% of the NRM. As for the latter, there is evidence that the positive polarity P remanence has very high coercivities.

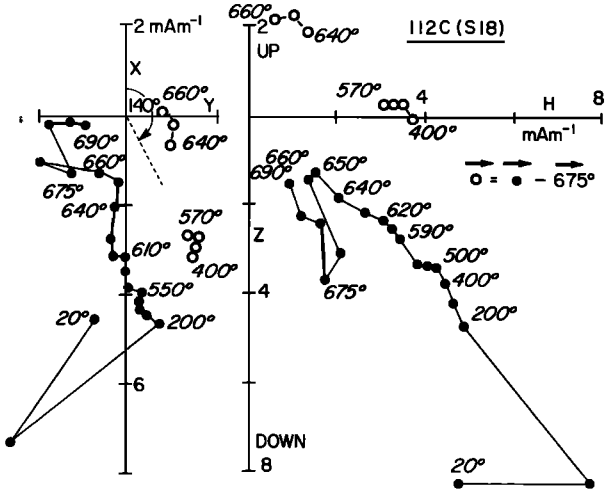


Fig. 3. Vector diagrams of thermal treatment showing the observed vector remaining in intermediate to steep positive inclinations, unlike specimen 153C (Figure 2) where the inclination is shallow in the 450° to 620°C range. Vector subtraction shows two features: (1) the similarity in the 400° to 570°C range between the sought vector (low inclination and a declination of 140°) and the observed vector of Figure 2, and (2) the similarity between the sought vectors in the high T_{UB} range (640° and upwards) in both instances. The 675°C vector (not the mean of the high T_{UB} vectors) was used for the vector subtraction. The choice of using one or the other procedure is dependent upon the intensity decay at high temperatures. The use of either procedure does not affect the argument qualitatively since the sought vector is an approximation as explained in the text. However, for increased accuracy in the determination of the +P direction, the mean direction of the 675° to 690°C vectors (251°, +63°; $\kappa = 84$; $\alpha_{95} = 10°$) was used for the compilation of the site 18 site mean (Table 1).

TABLE 2. Analysis of Directions Obtained After Treatment in the Range of 550°–630°C or 70–140 mT Before (R) and After (R') Vector Subtraction of the High Stability Remanence (P)

	R					$R' = \sum(\bar{r}_i - \bar{p}_i)$				
Site	N_3	D°, I°	κ	α_{95}	Pole Long., Lat.	N_4	D°, I°	κ	α_{95}	Pole Long., Lat.
					Group A					
29	10	162, +04	8	18	137°E, 40°N	10	154, −07	05	30	149°E, 43°N
28	10	155, +11	3	33	144°E, 35°N	10	144, −02	04	28	159°E, 36°N
27	9	172, +17	11	16	133°E, 36°N	9	147, −04	11	16	156°E, 38°N
26	12	125, +04	13	12	175°E, 23°N	11	108, −07	22	10	193°E, 15°N
25	11	116, −15	13	13	190°E, 23°N	11	111, −23	23	10	196°E, 24°N
24	5	157, −17	9	27	149°E, 49°N	5	141, −25	19	18	171°E, 44°N
Mean	6	148, +01	10	22	154°E, 36°N	6	135, −12	14	18	172°E, 34°N (9,18)†
					Group B					
18	11	178, +43	55	06	115°E, 20°N	11	175, +38	36	08	111°E, 23°N
19	13	155, +32	16	11	139°E, 24°N	13	153, +27	19	10	143°E, 26°N
20	8	155, +25	6	24	141°E, 27°N	8	162, +17	23	12	135°E, 34°N
21	15	179, +38	17	09	104°E, 24°N	2*	145, +33	...	...	...
22	14	188, +38	44	06	104°E, 23°N	14	178, +22	10	15	115°E, 33°N
23	6	160, +06	12	27	139°E, 39°N	4	139, −42	07	39	184°E, 51°N
Mean[1]	6	168, +31	20	15	126°E, 27°N	5	162, +14	06	35	135°E, 35°N
Mean[2]	5	170, +36	33	14	123°E, 24°N	4	167, +26	35	16	128°E, 30°N (10, 18)†

N_3 indicates the number of specimens yielding a shallowly inclined (r) direction. N_4 indicates the number of specimens yielding both p and r; R' has been computed by subtracting at the specimen level, p vector (Table 1) from the r vector, both p and r usually being the mean of 3 to 4 observations. Other symbols as per Table 1. Mean[1] and Mean[2] are the averages depending on whether site 23 is included or not. All directions given with respect to bedding (corrected for tilt).

*Not used for 'mean' because of small number of determinations.

†The semi-axes (δ_p, δ_m) of the oval of confidence at probability level of 0.05.

In some instances, the vector becomes less negatively inclined (Figure 5) which is probably attributable to a change in the intensity ratio of positive/negative polarities according to the evidence presented below.

AF plus thermal treatment. Although the early runs in high AF provide one with some evidence of a negatively inclined remanence, the difficulties encountered indicate that further runs would probably not have resulted in an accurate determination of that direction. The remaining specimens were therefore transferred to thermal treatment after the 100–140 mT treatment. Virtually no changes occur before the 400°–500°C treatment indicating that the remanence with T_{UB} in the 20° to 400°C range has coercivities below 140 mT (Figure 6). It is noted that the results are in good agreement with the straight (non-AF treated) thermal experiments where the movement towards low inclination occurs in the 20° to 400°C range. In accordance with the pattern followed in the thermal experiments, the vector again gradually steepens positively in the 400° to 650° range until it reaches and stablizes in the P direction. The intensity, however, is much less than in the non-AF treated specimens indicating that part of the remanence with high T_{UB} (up to 690°C?) has coercivities below 140 mT.

These observations show that each technique provides one with varied and valuable information about the magnetic characteristics of these beds. For example, comparison of results obtained from thermal, AF and AF plus thermal experiments shows that the high T_{UB} remanence is not directly and proportionally related to the high coercivity remanence but has a rather wide coercivity spectrum extending both above and below the 140 mT treatment. Findings of this kind testify to the desirability of performing extensive experimental work using all the presently available techniques in order to retrieve as much information as possible and improve our limited knowledge of remanence acquisition processes.

COMPILATION OF RESULTS

The P remanence is clearly defined and can be recovered by independent methods. Results obtained from the different treatments are in good agreement as shown in Figure 7. The chemical and AF results are slightly less well grouped than the thermal results and this reflects the fact that fewer determinations were obtained chemically or by AF (1–4/site) than thermally (4–8/site). In addition, the AF technique was less effective for the recovery of this remanence and failed to do so at 4 sites.

For each specimen, the directions were determined using methods described in detail by *Roy et al.* [1979]. Hence P was usually defined by averaging the last four vectors. For example, this yields a calculated direction for specimen 153°C (Figure 2) of 184°, +68° ($\kappa = 548$; $\alpha_{95} = 4°$); additional information is given in figure captions. Site-mean directions were calculated using results obtained from each of the techniques (Figure 7) and also combining all the determinations obtained from the various techniques. (A sample of the procedure is given later in Table 3 which again shows good agreement among those results.)

The site-mean directions and statistics of the combined P results are given in Table 1. At some sites, the number of observations (N_2) is less than the number of treated specimens (N_1) and this is attributable to failure to detect the P remanence, mainly in specimens treated in high AF (150 to 290 mT). As argued later, all the experimental results can be reconciled only by assuming that the remanence was acquired at and/or shortly after formation of the beds which evidently implies a prefolding remanence acquisition. Therefore, the di-

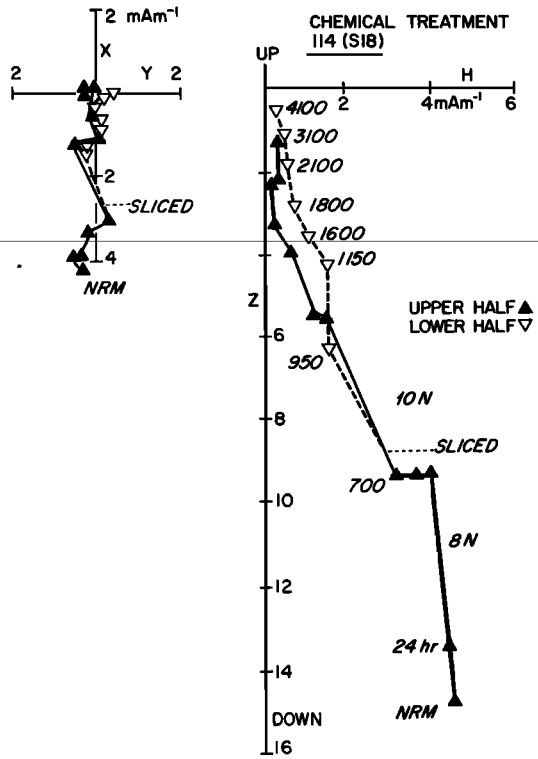


Fig. 4. Results obtained from chemical leaching experiments showing that the steeply inclined remanence *P* predominates at all times. The specimen was sliced after the 700 hour treatment and the two halves were subjected to further leaching up to 4100 hours. The agreement between the intensity decay curves of the two halves is remarkable. There are no indications of a *R* direction. The results also indicate that if some of the carriers were magnetized in the opposite polarity, they are more soluble in acid than the positive polarity remanence carriers. Slicing revealed that 70% of the specimen had lost its initial color. The large decrease in intensity from 700 to 950 hours is attributed to two factors: the increase in acid strength to 10 *N* and exposure of the center of the specimen to acid following slicing.

rections (in Tables 1 and 2 and Figures 2–8) are given after correction for tilt has been applied. As a matter of interest, without correcting for tilt, the group mean directions are 211°, +84° (pole 106°E, 35°S) for group A and 179°, +80° (pole 114°E, 26°S) for group B with comparable statistics.

The results obtained after treatment in the 550° to 630°C and 70 to 140 mT ranges are compiled in Table 2. If it is assumed that direction *R* is the result of a remanence acquired long after the acquisition of *P*, then an estimate of the field direction at that time requires the removal of the effect of the (higher T_{UB} and/or coercivity) *P* remanence. At least part of the bias of the *P* remanence can be removed vectorially to delineate a *R'* direction. Vector analysis was performed at the specimen level using methods described in Figures 2 and 3, for example, and the results (*R'*) with respect to bedding are also compiled in Table 2. Without correction for tilt the group mean directions are 142°, −12° for group A and 160°, +37° for group B.

It is most important to note that vector subtractions yield only an approximation of the sought vector because the remanence used to represent the subtracted vector may have been reduced between the two treatments used for the analysis [*Roy and Park*, 1974]. For example, Figure 3 shows the results obtained by subtracting the *P* remanence vector observed at 675°C. However, the T_{UB} spectrum of *P* may extend below 675°C to perhaps 640°C. If so, the intensity of the subtracted remanence is slightly low; it can be readily seen on the vector diagram that subtraction of a larger remanence would take the 640°C difference vector closer to antiparallelism with the 675°C vector. Likewise for the AF plus thermal results, the positive *P* remanence vector with high T_{UB} was used for the vector subtraction from the 70 to 140 mT vectors. However, part of remanence *P*, having coercivity in that range, has already been removed so that the intensity is too low. If the full intensity of the original *P* remanence remained, all difference vectors would be more closely antiparallel to *P*. Therefore the directions given in Table 2 and the figures are maximum departures from the −*P* direction.

THE MAGNETIC HISTORY

Although there is a remote possibility that the *R'* direction represents an overprint acquired long after deposition (interpretation *A*), the considerable information gathered through the use of the various techniques is much more consistent with the hypothesis that all the remanence (with $T_{UB} > 100$°C and coercivities > 5 mT) was acquired during a brief period spanning a polarity reversal which probably occurred during formation of the beds (interpretation *B*). Before assessing the validity of the two interpretations, some of the information about red bed remanence which has come to the fore in the

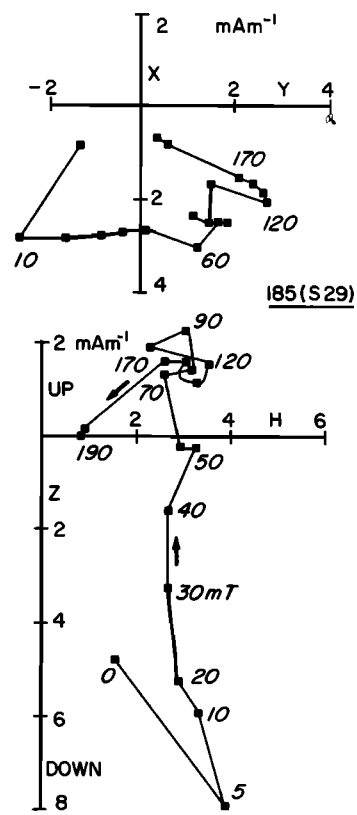


Fig. 5. Diagrams of vector changes during AF treatment. The vector moves to a negative inclination at 70 mT where it remains but with irregular changes until 170 mT. It then apparently attempts to return to a positive polarity. From 200 to 290 mT the changes are very irregular with intensities much larger than the 0.9 mA m^{-1} at 190 mT. They are rejected according to criterion 2. Note the similarity of the horizontal component to that of the thermally treated specimen (Figure 2).

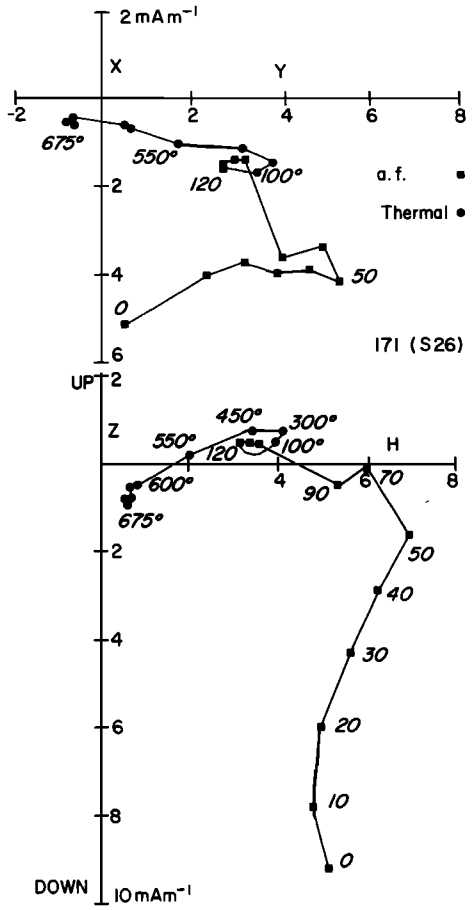


Fig. 6. Vector changes during AF treatment followed by thermal treatment. Again the vector reaches a negative inclination upon AF treatment. However part (the intensity being much smaller than those of the thermally treated speciments—Figures 2 and 3) of the positive polarity remanence is uncovered by subsequent thermal treatment.

last few years through other detailed investigations, usually involving a combination of thermal, chemical and/or AF experiments, is briefly reviewed.

Other studies. In a number of Precambrian to Carboniferous red bed studies (*French and Van der Voo* [1977], *Morris* [1977], *Van der Voo and French* [1977], and *Van Alstine and Gillett* [1979], to name a few), it has been found that the apparent complexity of the results could be attributed to a single cause: the occurrence of a field reversal(s) during remanence acquisition. Concurrently, several (seven) other investigations (for references, see *Roy and Morris* [1981]) have shown that, for many red beds, the initial remanence acquisition of most red beds is a lengthy process operating during the formation of the beds (from deposition to lithification) and comprises different phases (or types) of magnetization. Not all phases and their carriers have characteristics identical in all aspects—T_{UB} and coercivity spectra and solubility in acid—so that through the use of different techniques it is often possible to decompose the remanence and gain some insight into the acquisition process [*Roy*, 1977; *Henry*, 1979]. Thus, in some instances, it has been demonstrated that the process is long enough to space field reversal time scales (a few thousand years) meaning that specimens can carry dual polarity remanences acquired partly before and partly after a reversal. An upper time limit for the process has not been established with precision and probably varies from formation to formation. However, when it can be shown that all phases yield approximately the same direction (irrespective of polarity), it can be argued that the whole remanence acquisition process was relatively rapid and/or no appreciable polar wander occurred during the process.

Although remanence acquisition may be a continuous process comprising several intergrading phases, *Roy and Park* [1974] explained their results using a simplified three-phase model: the first phase being the mechanical orientation of ferrimagnetic mineral particles during deposition, the second

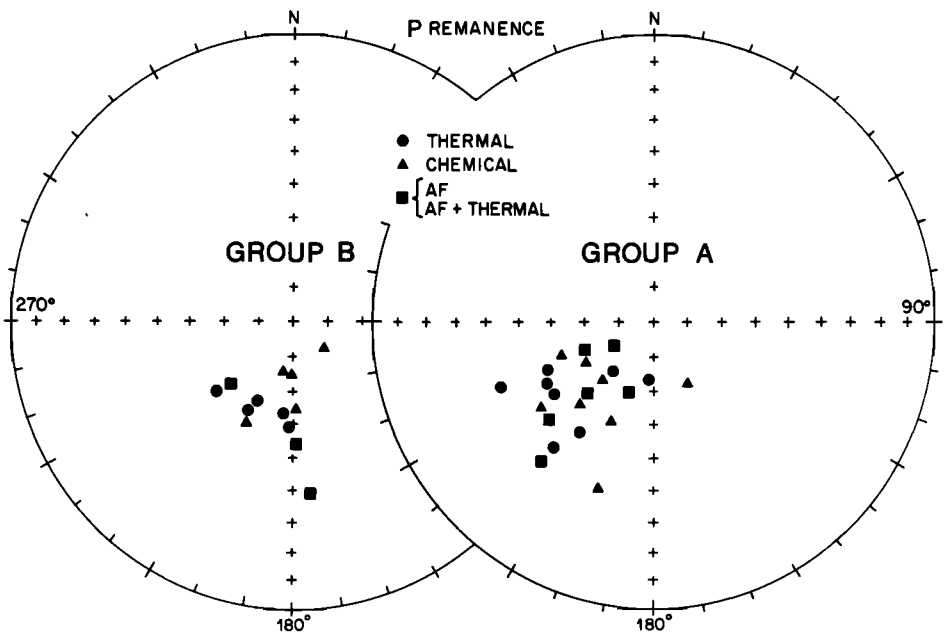


Fig. 7. Site mean directions of the *P* remanence as determined by the different techniques. The dots represent the mean of 4 to 8 observations while the triangles and the square are the means of 1 to 4 observations. At 4 sites, the remanence could not be recovered by AF treatment. Site 21 is not included.

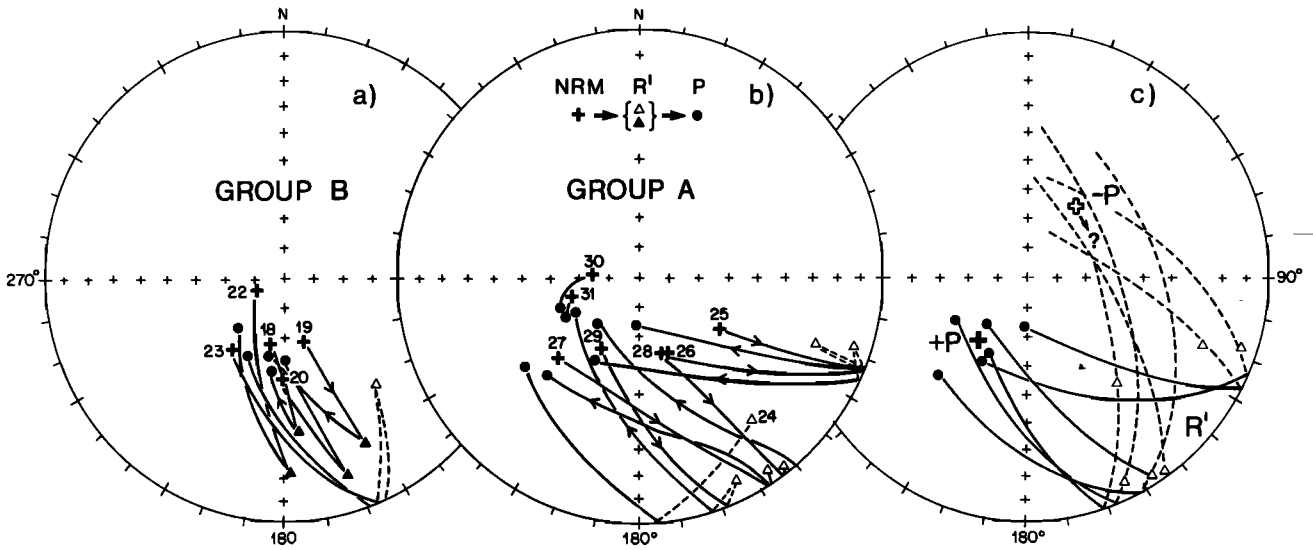


Fig. 8. Representation on equal area stereonets of results compiled in Tables 1 and 2, open symbols indicating negative inclinations and all other symbols positive inclinations. In Figures 8*a* and 8*b* the site mean directions of the NRM, R' and P are joined (not necessarily along great circles to facilitate legibility). The general agreement between the NRM and P results indicates that most of the NRM was acquired in the P direction. In Figure 8*c* great circles passing through the P and R' directions are shown. To avoid cluttering and improve resolution, only the sites with the most divergent R' directions are shown. The circles intersect near only two locations P and $-P$ representing the two polarities of the average (216°, +62°). It is suggested that failure to find another convergence point indicates that R' is not a distinct remanence but the resultant of a dual polarity remanence. Although difficult to assess, the greatest concentration of intersecting circles appears to be slightly to the south (-east?) of the plotted $-P$ which would indicate a sub-antiparallel direction. This would account for the R' being confined to the second quadrant.

being an hematite growth by authigenesis and the third being produced during compaction and cementation during lithification. The three-phase model has been found to be applicable to other red bed studies (for references, see *Roy and Lapointe* [1978]) and is adopted here to test the validity of interpretation *B*.

Present study. The similarity between the P and the NRM directions, especially for group B (Table 1), gives a first indication that the whole process took place in a relatively short time. Those directions probably represent (with different accuracies) the directions of the field at the beginning and the end, respectively, of the remanence acquisition process. In accordance with other studies, both the chemical and thermal experiments indicate that the P remanence belongs to the early phase(s); its carriers are very resistant to acid leaching, often a characteristic of first-phase remanence, and its T_{UB} is extremely high (660°–690°C), characteristic of first- and second-phase remanences. It is concluded that the P remanence was acquired during the first phase, and possibly partly during the second phase since the chemical experiments may not have been totally effective as pointed out earlier. Likewise, both the chemical and thermal experiments indicate that the NRM is carried predominantly by the third phase as indicated by the large decrease of intensity upon disappearance of the cement (Figure 4) and the corresponding T_{UB} range from 20° to 500°C (Figures 2 and 3). Therefore the NRM directions can be regarded as a rough estimate of the field direction during the acquisition of the last phase which occurred during cementation.

Conversely, the fact that remanence P has T_{UB} from 20° to 500°C and 650° to 690°C renders interpretation *A* very laborious. It is difficult to conceive how an overprint remanence could be limited to a T_{UB} window of 500° to 650°C without affecting the lower range of T_{UB}. One would also have to find an explanation for the 500° to 650°C gap in the T_{UB} spectrum of P. Furthermore one would expect a later overprint to have affected both groups A and B contemporaneously to produce a common R' direction which is not the case (Table 2 and Figure 8). In addition, sites 30 and 31 would somehow have escaped this later event. Interpretation *A* is therefore considered invalid.

Consequently, the R' direction is regarded as the result of remanences produced during the initial process. It is, however, unlikely that it represents the direction of the field. Figure 8 shows that the large changes in direction would require an excessively rapid (up to 180° return) apparent polar movement which by coincidence would begin and terminate in approximately the same location!

A simpler explanation is that part of the remanence was acquired in the opposite polarity ($-P$) owing to the occurrence of a field reversal at that time. There is good evidence for the hypothesis. In some specimens, the removal of the high T_{UB} P by vector subtraction clearly indicates the existence of a steep negative inclination nearly antiparallel to the P direction (Figures 2 and 3). Negative inclinations are also revealed directly in a number of specimens especially during AF treatment (Figure 5). There is also the fact that the great circles connecting P and R' converge in two locations only (P and $-P$; Figure 8*c*). Finally a model based on this hypothesis can explain and is compatible with all the results obtained from the different techniques, each revealing different characteristics (Figure 9).

A Hypothetical Model

The model consists of decomposing the observed remanence into three phases acquired successively, the first and

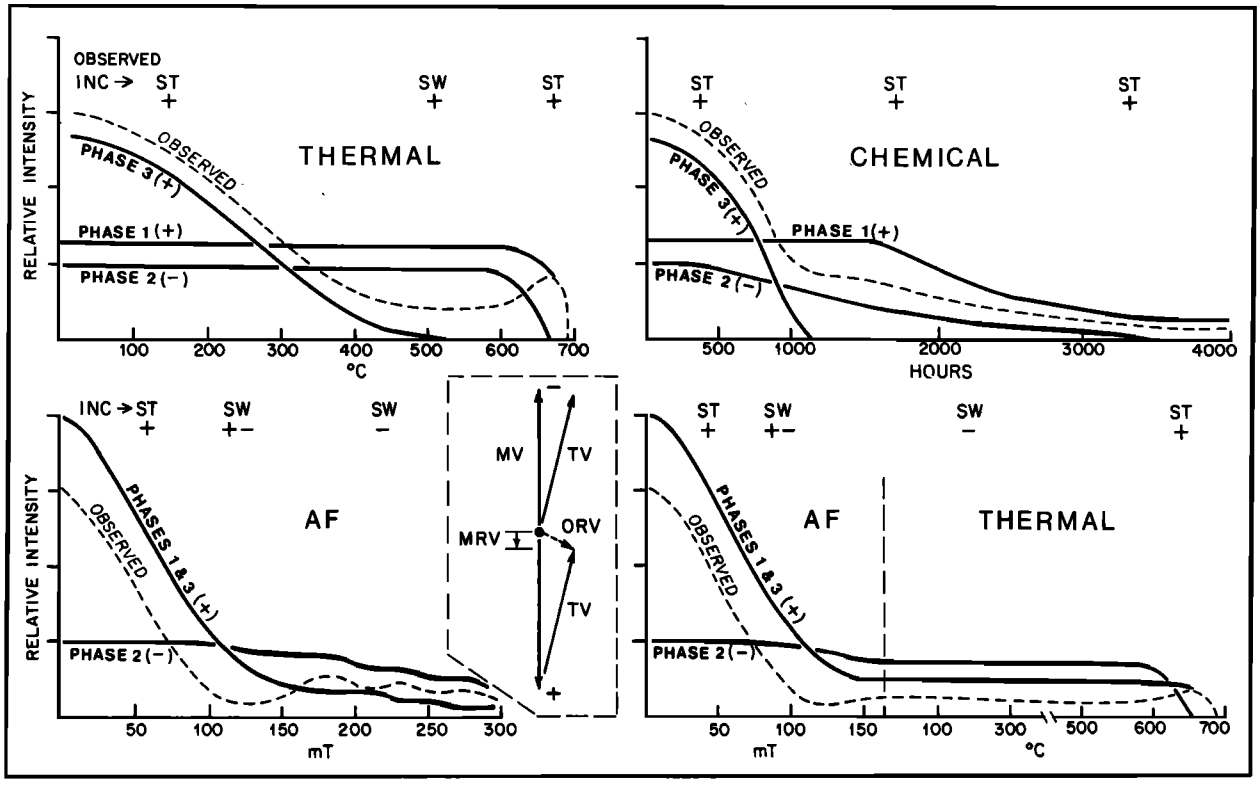


Fig. 9. Generalized intensity decay curves closely depicting the remanence behavior 'observed' in this study during thermal, chemical, AF and AF plus thermal treatments. Based on this information and supporting evidence from other red bed investigations (see text), the remanence is decomposed into three phases nearly parallel in direction (along P), phases 1 and 2 being positive and phase 3 negative. The inclination (positive and/or negative) observed during treatment is indicated: ST for steep and SW for shallow. The only discrepancy between this model and the observations is that in shallow inclinations the observed vector (R') is sometimes larger than the resultant vector drawn in the model. The explanation for this is that the model is based on exactly antiparallel vectors while in reality the vectors are more or less off antiparallelism. The effect is illustrated in the vector diagram in the center where the true vectors (TV) are 13° off antiparallelism with MV representing the model vectors (the unlabelled positive vector representing both, the true and model vectors). MRV and ORV are the model and observed resultant vectors, respectively. The ORV/MRV ratio is larger than 2.

third being positive (parallel to P) and the second being negative (antiparallel ($-P$)) (Figure 9). The decay curves of the 'observed' remanence are based on a generalization of the observations, normalized to the NRM intensities. The hypothetical intensity ratios closely approximate the general trend observed during treatment except that in the part of the experiments where shallow inclinations are encountered, the intensity observed is sometimes larger than the modelled intensities (see Figure 9 caption for explanation). Variations of this model are of course possible since some details remain unknown, e.g., the relative intensities of phases 1 and 2 at 20°C. However, slight changes in slope of either or both phases in the 20° to 600°C range would not invalidate the model.

The thermal model. This model is easily constructed since much information about the T_{UB} spectra of the different phases is available from other red bed studies. The positive polarity phases 1 and 3 account for the steeply inclined NRM. As the phase 3 remanence is removed, the ratio of positive/negative polarity remanences diminishes and the resultant vector yields a low inclination (Figure 3; 400° to 650°C range). In the 650° to 690°C range, the steepening of the vector into the positive polarity indicates that the phase 2 remanence has slightly lower T_{UB} than the phase 1 remanence. This is interesting because it is the first evidence in the literature that phases 1 and 2 can be separated thermally, although unpublished experiments performed on other red bed formations have indicated that the two phases could be separated with the use of small (2°–5°C) incremental steps (W. A. Morris, private communication, 1980).

The smooth transition from shallow to steep inclination often observed in the high T_{UB} range (Figure 2) has been translated in the model by gradual decrease of both phases 1 and 2 in the 600° to 690°C range. This can explain why the $-P$ direction is not always recovered even with vector analysis. The model illustrates that in a vector subtraction of say 600°–690°C, the subtracted ('observed') vector is about equal in intensity to the intensity contributed by the negative polarity of the 'observed' 600°C vector. Therefore any offset from antiparallelism of the two vectors can produce large departures from the $-P$ direction towards low inclination as shown in the vector diagram (Figure 9). The fact that the declinations are limited to the second quadrant could be indicative that the $-P$ direction is not exactly antiparallel to the P direction but deflected towards the southeast by an unknown amount (Figure 8).

The chemical model. This model shows that the combined phases 1 and 3 predominate over the phase 2 remanence at all

times. Appreciable departures from the P direction (or movement towards the $-P$ direction) are prevented owing to the slow penetration of acid which starts attacking the phase 2 carriers before complete removal of the phase 3 carriers.

The AF model. Information about the respective coercivity spectra of the different phases is limited. Two previous studies comparing thermal, chemical and AF results indicate that phase 3 has a coercivity spectrum between 0 and the 200 to 250 mT range while both phases 1 and 2 have a substantial portion of coercivities above 250 mT [*Roy,* 1977; *Roy and Robertson,* 1978]. This is in agreement with the results obtained in this investigation. However, because the lower portion (<200 mT) of the coercivity spectrum of phase 1 cannot be determined, the relative contributions of phase 1 and 3 to that part of the intensity decay curve are unknown. Therefore, phases 1 and 3 are not decomposed but represented by a single curve, although it is probable that most of the large decay up to 150 mT is attributable to phase 3 while the rest of the curve represents almost exclusively phase 1.

The combined phases 1 and 3 first clearly predominate over phase 2 as shown by the steeply inclined NRM. Then the vector usually becomes negatively inclined in the 70 to 140 mT range (Figures 5 and 6) and remains so until the end of the treatment (290 mT) indicating that the phase 2 has a higher portion of its remanence with coercivities above 140 mT than phase 1 has. Tendencies towards the positive polarity (Figure 5) substantiate the inference that some of the phase 1 remanence has coercivities in that range. The overlapping coercivity spectra of the two phases also account for the variations in intensity from 150 to 290 mT.

The AF plus thermal model. This technique provides one with additional information about the coercivities of phases 1 and 2. That some of the phase 1 remanence has coercivities >140 mT is clearly indicated by the recovery of the P vector in subsequent thermal treatment. As in the thermal model, the phase 2 remanence is shown to decrease in the 600° to 650°C range to reveal the phase 1 remanence with its higher T_{UB}. The intensity is clearly smaller (compare Figure 6 with Figures 2 and 3) indicating that part of the phase 1 remanence has coercivities <140 mT.

Examination of polished sections shows the presence of hematite grains and clouds of unidentified fine red particles beyond the resolving power of the microscope. In the leached portions only a minute amount of possible magnetic carriers is present. No magnetite grains could be positively identified and it is assumed based on the low NRM intensities (≈ 20 mA m^{-1}) that hematite is the major if not the only magnetic carrier.

Since all the results are consistent with the hypothesis of a single dual polarity remanence, interpretation B is accepted. It is noted that if fewer techniques had been used, the validity of the hypothesis would have been much more difficult to demonstrate. Most importantly, if the experimental work had been limited to thermal treatment up to 650°C, the R direction (Table 2) could possibly have been mistaken for the initial remanence (with the NRM as a recent remagnetization?). This false interpretation might have appeared reasonable since the poles (especially that of group B) are in reasonable agreement with other Siluro-Devonian poles as shown later.

An F distribution test for comparison of mean directions [*Watson,* 1956] indicates that, for both the NRM and the P results, respectively, the group A and group B mean directions are not significantly different at the 95% level of confidence so that the average (Table 1) is regarded as a more accurate estimate of the field than the group means. The test also indicates that the NRM and P averages are not significantly different at the 95% level, in accord with the hypothesis that most of the remanence was acquired in the P direction (irrespective of polarity). Of the two averages, P (direction: 216°, +62°; pole: 087°E, 05°S, Table 1) is considered a better estimate of the field direction than the NRM (direction: 192°, +68; pole: 105°E, 07°S), since it has been experimentally isolated while the NRM, which comprises opposite polarity phases, could be biased owing to sub-antiparallelism of the two polarities as discussed previously (e.g., captions of Figures 8 and 9).

THE INTRUSION

One site (32) was sampled from the intrusion, the exposed part of which is 1 m wide. No contact could be found with the adjacent group A sedimentary unit because of overburden. The samples at the nearest sites 30 and 31, 3 and 4 m away, respectively, do not appear to have been affected by the intrusion. They do not exhibit any of the characteristics of a TRM (thermoremanent magnetization) produced by the intrusion of a nearby igneous body; e.g., the NRM intensities are within the range of intensities for the group and not larger by factors of 10 to 24 as found from comparison of samples of similar composition (argillites) near and far from a contact [*Roy and Lapointe,* 1976, p. 761].

Since only one site could be sampled, the results (8 specimens) are of limited value but nonetheless worth reporting because of their similarity with previous results. The NRM (mean intensity = 1182 mA m^{-1}) directions are scattered: direction 338°, +18°; $\kappa = 1.6$. Two remanences of high stability are present: (1) Z_m with T_{UB} of 500°–600°C and coercivities 20–50 mT (intensity = 90 mA m^{-1}; direction = 027°, −25°; $\kappa = 26$; $\alpha_{95} = 11°$) and (2) Z_h with T_{UB} of 600°–690°C determined with and without prior AF treatment (intensity = 51 mA m^{-1}; direction 325°, −49°; $\kappa = 39$; $\alpha_{95} = 9°$).

Z_m which could be carried by magnetite or titanohematite is comparable in direction to that of group C (described below) and one (018°, −27°) of the two directions obtained from the St. George pluton. Z_h, probably carried by hematite, is comparable in direction to that of another intrusion (site 13: 329°–54°) of the Mascarene Group [*Roy et al.,* 1979]. The Z_m and Z_h directions yield poles at 083°E, 26°N and 143°E, 09°N, respectively.

GROUP C

Two directions of remanence (M and H) are also present in specimens of this group. However, both the directions and the magnetic characteristics are quite different from those of groups A and B.

First, the NRM intensities (≈ 101 mA m^{-1}) of these graywackes are about 10 times larger than those of the siltstones. At two sites the directions (with respect to bedding) are fairly well grouped (≈ 027°, −02°) while at the other sites they are scattered, all with a positive inclination ranging from intermediate to high but not centered about the present earth's field direction as shown in a later figure (Figure 13). Again the specimens were subdivided and subjected to thermal, chemical and AF treatments. The results obtained from the three treatments are in good agreement, as shown in Table 3, with the vectors gradually moving to a negatively inclined direction with a marked improvement in grouping (larger κ). That direction is usually reached at about 300°C (Figure 10), 30

TABLE 3. Comparison of Results Obtained From Thermal, Chemical and AF Treatments; Group C; M^1 Corrected*

	NRM				Thermal			Chemical			AF			Combined	
Site	N_a	Int, mA m^{-1}	D°, I°	κ	N_t	D°, I°	κ	N_c	D°, I°	κ	N_f	D°, I°	κ	D°, I°	κ
14	10	70	0.88, +78	7	4	023, −30	19	3	031, −36	13	3	034, −05	165	029, −24	13
15	10	67	084, +62	10	4	030, −20	56	2	026, −31	···	4	016, −20	51	024, −22	30
16	7	103	022, −03	14	4	024, −24	108	2	022, −30	···	1	010, −30	···	022, −26	96
17	5	163	032, −01	25	3	025, −24	70	1	029, −20	···	1	018, −27	···	025, −24	127
Mean	4	101	···	···	4	026, −24	262	4	027, −29	117	4	020, −21	30	025, −24	666

The number of specimens (N_a) treated thermally (N_t), chemically (N_c) or by AF (N_f) is indicated. Int is the arithmetic mean intensity.
*See Table 4 for definition of M^1.

mT (Figure 11) or 700–950 hours (Figure 12) where it essentially remains until about 550°C, 60 mT and 4500 hours (direction M).

Further thermal treatment reveals the presence of a southerly and positively inclined remanence (H) with T_{UB} of 600° to 670° C (Figure 10). That remanence is also detected in AF treatment of 150 to 290 mT, except in one instance. However, as with group A and B specimens, irregular changes occur from treatment to treatment so that most specimens were transferred to a thermal run after the 100–150 mT treatment. In all but two specimens, an H remanence with $T_{UB} \geq 650$°C is uncovered (Figure 11). Remanence H could not be uncovered in any of the chemically treated specimens (Figure 12). All specimens gradually disintegrate from the 3100 hour to a subsequent 5500 hour treatment so that the relative resistances to acid leaching of the M and H carriers cannot be established. Since the vector remains in the M direction until disintegration, it could be argued that the H carriers are destroyed first. However, it is noted that the intensities at the last measurements are in the 30 to 50 mA m^{-1} range and therefore larger than the apparent intensities of the H remanence (e.g., compare Figure 12 with Figures 10 and 11). It is therefore possible for the M carriers to be less resistant and yet have a remaining remanence larger than the H remanence. Thus, observations based on decaying vectors are inconclusive. However, based on examination of polished sections (described below), it appears that the H carriers are not more and probably less resistant to acid leaching than the M carriers.

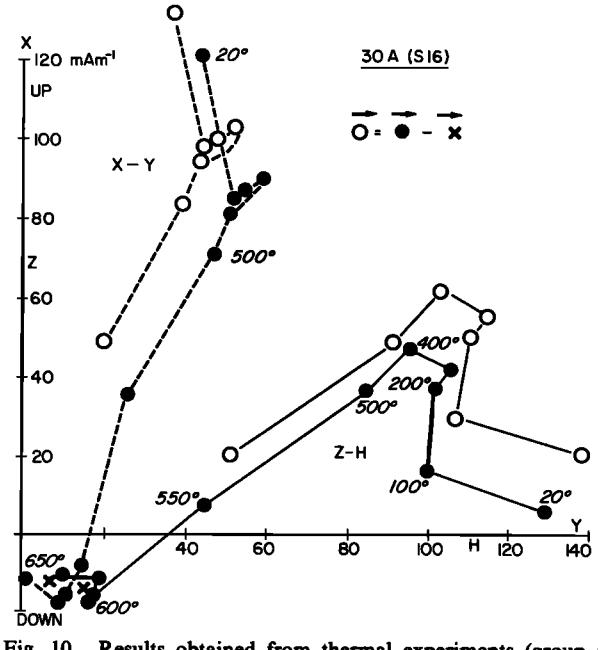


Fig. 10. Results obtained from thermal experiments (group C) identifying two remanences (solid dots); M with T_{UB} of 300° to 500°C and H with T_{UB} of 600° to 650°C. The open circles are the result of removing the effect of the observed H remanence, in order to recover M'; the declination is reduced and the inclination increased (negatively) as shown in Table 4. The 200° to 550°C M' vectors yield a mean direction of 025°, −25° with a $\kappa = 561$ and $\alpha_{95} = 3$°. Correction for tilt applied.

Although H cannot be detected chemically, its presence is clearly established from thermal, AF and AF plus thermal experiments. Except for 3 specimens whose directions of remanence remain in the M direction, the directions of all other specimens (21) so treated are shown in Figure 13. The site means are given in Table 4. The grouping is only fair. This can be attributed to at least two causes: (1) the large intensity ratio of M/H making H difficult to determine accurately and (2) the inclusion in the compilation of all results, particularly the 3 divergent and questionably reliable directions with low inclinations shown in Figure 13.

As a result of the underlying H remanence, the observed vector m is the sum of the m' and h vectors, respectively, representing M, M' and H at the specimen level. The effect of h is removed, at least partly, by subtracting the average of the h vectors observed at high T_{UB} (or after AF treatment in high fields) as illustrated in Figures 10 and 11, again realizing that the initial intensity of h could possibly be larger. The fact that the subtracted vectors do not always decrease towards the origin (Figures 10 and 11) indicates that m' and h have, to a certain extent, overlapping T_{UB} and coercivities. Conversely, because of the large increase in precision κ (Table 4; compare M' with M), it would appear that most of the T_{UB} and coercivities of the h remanence is above 600°C and 50 mT. We therefore regard the average M' as a reasonable estimate of the field at the time of acquisition of that remanence.

AGE OF REMANENCE

The exact ages and sequences of acquisition of the M' and H remanences are not easily established. The rocks are from undifferentiated Silurian strata which according to *Ruitenberg* [1968] were subjected to slight metamorphism during Early to Middle Devonian. Therefore were one or both remanences acquired before or after tilting of the beds? Compilation of results with and without correction for tilt (Table 4) shows poles all lying in the general area of the few published Siluro-Devonian poles (Figure 14) so that none of the alternatives can be

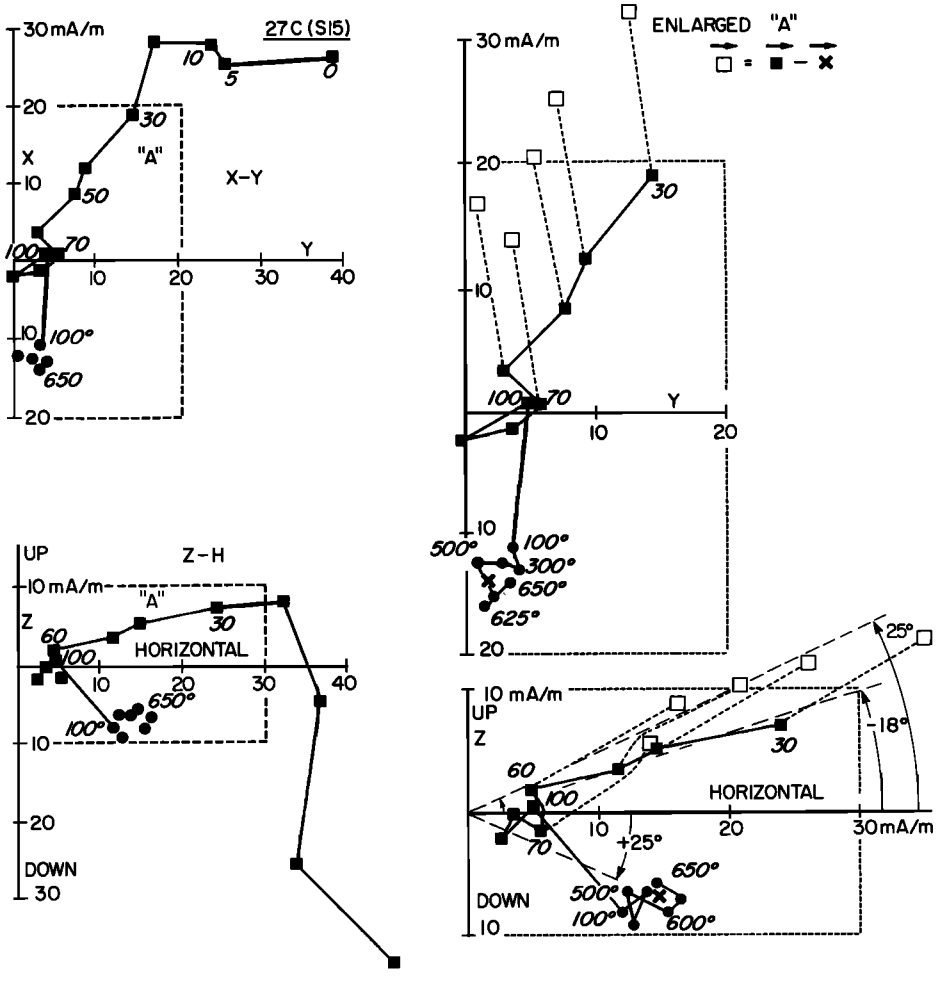


Fig. 11. Results obtained from two-stage treatment. The M direction is revealed by AF treatment of 20–60 mT. The presence of another remanence with higher coercivities is already signified here by the changes to positive polarity and southerly directions in the 70 to 100 mT range (also confirmed in other specimens treated in the 150 to 290 mT range). It is also confirmed here by the following thermal treatment which defined H more accurately. The M' direction (open squares) is determined by vector subtraction of the H remanence. The 30 to 70 mT vectors yield a mean M' direction of 015°, −25° with a $\kappa = 144$ and $\alpha_{95} = 6°$. Correction for tilt applied.

rejected on purely paleomagnetic grounds, that is, by using the dubious practice of assessing the reliability of a pole according to its agreement (or disagreement) with an 'established' polar path.

Polished sections of thermally, AF and chemically (300 and 3100 hours) treated specimens were examined in reflected light. They reveal the presence of both magnetite and hematite grains (in various ratios averaging about 4/1) in accordance with the experimental results, the magnetite accounting for the large NRM's and the hematite for the T_{UB} as high as 670°C. M' is therefore associated with the magnetite and H with the hematite. This is supported by the fact that no traces of hematite can be found in the chemically leached (3100 hours) specimens which fail to reveal the H remanence. Although the amount of magnetite is also greatly reduced, it would appear that the hematite grains are leached away before the magnetite grains are. Traces of slight mineralogical changes are observed in thermally treated (670°C) specimens. However, this is believed to be a reflection of alteration produced during laboratory heating (in air). In the AF treated

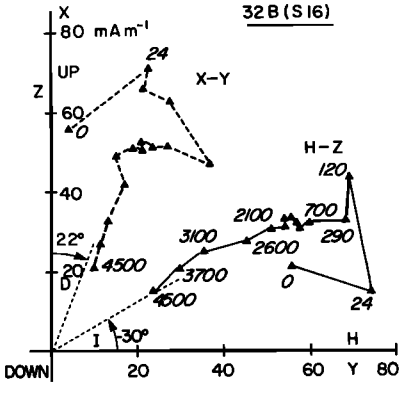


Fig. 12. Results obtained from chemical leaching experiments, 8 N acid strength used for first 700 hours and 10 N thereafter, until disintegration at 5500 hours. Only the M' remanence is revealed. However the intensity remaining at 4500 hours is larger than the apparent intensity of the H remanence (Figures 10 and 11). Correction for tilt applied.

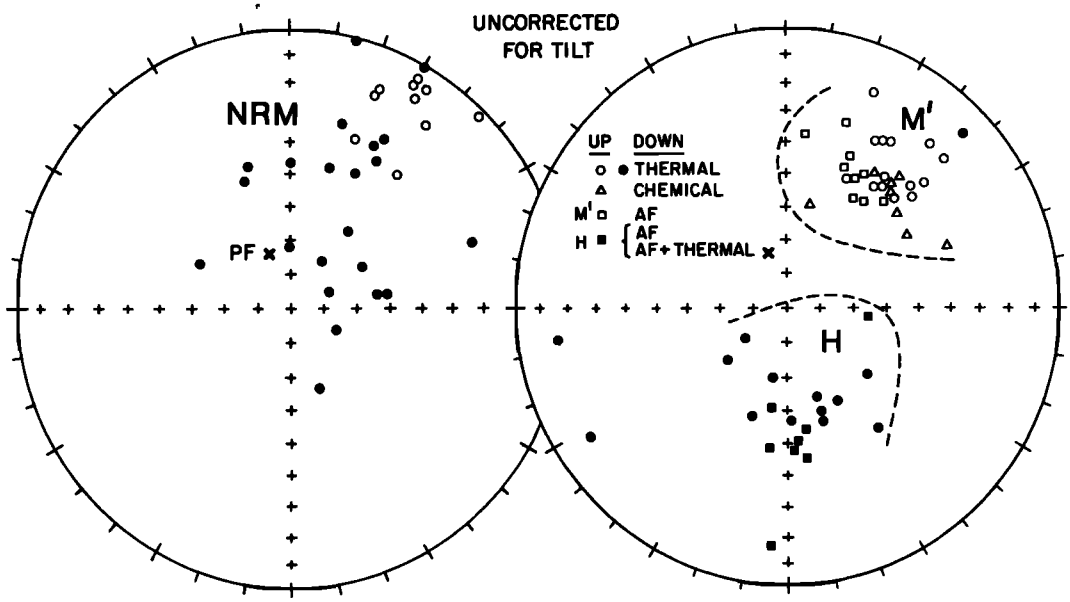


Fig. 13. All group C directions before (NRM) and after treatment as identified. In M', the directions from all 32 treated specimen are shown. In H, the directions obtained from 21 of 24 thermally and/or treated specimens are shown. A remanence other than M' could not be detected in the other 3 or any of the chemically treated specimens. The directions are plotted uncorrected for tilt for comparison with the earth's field direction. Since the attitudes of the beds at the 4 sites do not differ greatly, the distribution of directions after correction for tilt is reasonably similar to the one shown but centered on 025°, −24° instead of 038°, −40° for M' and 175°, +33° instead of 182°, +58° for H.

and the unleached part of the chemically (300 hours) treated specimens, the magnetite particles are usually sharply defined without any sign of alteration and appear to be of detrital origin. Some of them, however, are partially or totally rimmed by hematite. There is also a minor fraction of particles totally composed of hematite. From these observations, it is concluded that the magnetite is the carrier of a DRM (detrital remanent magnetization) [*Verosub*, 1977]. It is impossible to determine if the homogeneous hematite particles were incorporated into the host rock to become DRM carriers or formed in situ during diagenesis to become CRM (chemical remanent magnetization) carriers. The hematite surrounding the magnetite particles is, however, probably a CRM carrier produced by alteration during diagenesis.

It is suggested that both M' and H were acquired prior to tilting and that M' corrected (025°, −24°) is representative of the field at the time of deposition. However, the significance of the difference (≈ 150°) between the M' and H directions cannot be properly assessed. Although the field produced by the magnetite could have caused the hematite to become magnetized in the opposite polarity, the preferred explanation is that a field reversal occurred between the acquisition times of the two remanences. If so, then the difference between the M' (085°E, 28°N) and the H (118°E, 26°N) poles might be indicative of an apparent polar displacement of some 33° in Silurian time. However, the departure from antiparallelism of the two remanences is possibly attributable to the lack of precision in determining the H direction as shown in Figure 13 and the relatively large α_{95} of 20° (Table 4). Owing to poor precision, H is not considered an accurate indicator of the field and consequently the evidence for an eastward polar displacement shortly after deposition of these strata is regarded as unsatisfactory and in need of further support from future studies.

DISCUSSION

Although the results of this investigation contribute valuable data towards the eventual unravelling of the tectonic history of the eastern seaboard and the Appalachians, they also confirm the complexities involved in attempting to redeem history by paleomagnetic means as evidenced by the wide scatter of Silurian and Devonian poles obtained so far (Figure 14).

A concentration of poles near longitude 120°E and latitude 20° to 50°N is clearly evident and it might be argued that the widely divergent poles (in south latitudes) are aberrant poles perhaps attributable to local rotation. However, it is noted that poles $B(R)$ and $C(H)$ are also found in that concentration. Pole $B(R)$ is derived from a direction obtained after thermal and AF treatments of 630°C and 140 mT, and which is shown to represent either an overprint or more likely an apparent remanence. Pole $C(H)$ is derived from a remanence with T_{UB} of 600° to 670°C and coercivities of 100 to 200 mT, but its significance as a field direction indicator is of questionable reliability. Several of the other poles belonging to the concentration have been obtained after less experimental work and the possibility that some of them do not represent the field direction at the time of formation is conceivable.

The results obtained in this investigation support the suggestion that substantial apparent polar movement took place in Siluro-Devonian times. Pole P indicates that the pole was near the equator in Pembroke time. This is in accord with preliminary results (pole 57) obtained from 3 Mascarene Group intrusives [*Roy et al.*, 1979]. Pole $C(M')$ together with poles 53, 55 and 58 indicate that at sometime in the Silurian, the pole was near latitude 30°N. The rock age of the group C sites becomes very important. According to *Cumming* [1967] and *Ruitenberg* [1968], these strata are of Waweig age meaning that

TABLE 4. Analysis of the Directions of Remanence Obtained From 4 Sites at Pagans Cove, Group C

		M				M′				H					
		Uncorrected			Corrected			Uncorrected			Corrected			Uncorrected			Corrected		
Site	N_a	D°, I°	κ	α_{95}	D°, I°	κ	α_{95}	D°, I°	κ	α_{95}	D°, I°	κ	α_{95}	N_b	D°, I°	κ	α_{95}	D°, I°	κ	α_{95}
14	10	046, −34	6	22	036, −17			041, −41	13	14	029, −24			6	185, +61	33	12	179, +37		
15	10	043, −30	9	17	034, −16			035, −38	30	09	024, −22			8	202, +49	6	24	191, +29		
16	7	038, −38	57	08	024, −24			036, −39	96	06	022, −26			4	195, +52	6	41	182, +32		
17	5	040, −37	63	10	027, −19			041, −42	127	07	025, −24			3	135, +59	48	18	146, +31		
Mean	4	042, −35	311	05	030, −19	164	07	038, −40	702	03	025, −24	666	4	4	182, +58	22	20	175, +33	23	20
Pole		071E, 15N			078E, 28N			077E, 14N			085E, 28N				111E, 06N			118E, 26N		
		(4°, 6°)*			(4°, 7°)*			(2°, 4°)*			(2°, 4°)*				(22°, 30°)*			(16°, 23°)*		

The directions are given with respect to horizontal (uncorrected for tilt) and with respect to bedding (corrected for tilt). M is the vector sum of remanence 'm' remaining after treatment at 550°C, 50 mT or 4500 hours in acid. H is the vector sum of remanence 'h' remaining after treatment at high temperature (600°–670°C), high AF (100–290 mT) or AF (50–150 mT) following by thermal treatment in the 600°–670°C range. M' is $\sum (m - h)$; when h could not be detected, it was assumed to be equal to zero. Results obtained from the three different treatments are given in Table 3 using M' corrected as an example. N_a is the number of specimens treated and yielding a 'm' remanence, and N_b the number of specimens yielding a 'h' remanence.

*The semi-axes (δp, δm) of the oval of confidence at probability level of 0.05.

during the Upper Silurian the pole relative to southwestern New Brunswick moved southwards (option *a*) by some 30°. This is apparently contrary to the evidence presented by pole Z_m of site 32 sampled from an intrusion believed to be younger than the group A (Pembroke Formation) sedimentary unit which would require a northward movement (option *b*). Normally not much weight should be given to Z_m since it is based on a single site. However both Z_m and $C(M')$ are in good agreement with a pole (81) derived from the lowermost Devonian St. George pluton. Does this indicate that the group C strata and the intrusion at site 32 are lowermost Devonian? It cannot be ascertained at the moment since the St. George pluton yielded two poles (80 and 81) and it could not be established which of the two was derived from the remanence acquired at the time of emplacement of the pluton. Assigning the initial remanence to pole 81 favors option *b* while such an assignment to pole 80 favors option *a*. The foregoing is a good example of the consequences of attempting to date poles according to their locations. Apparently only two options are open. This is not really so. The 'agreement' does not necessarily mean that the remanences were acquired contemporaneously. The remanences yielding the Z_m and/or 81 poles of indeterminate ages could have been acquired long after the $C(M')$ remanence; the agreement simply indicating that the apparent pole either remained in the same location during that span of time or came back to approximately the same location after executing some kind of excursion. Such ambiguities serve to illustrate the difficulty in, and importance of, ascertaining both the rock age and the remanence acquisition age.

In this respect, it is worth dwelling on the interesting suggestion of *Kent and Opdyke* [1978] and *Van der Voo et al.* [1979] that the northeastern seaboard was a terrain (area A) detached from cratonic North America during part of the Paleozoic. The hypothesis is partly based on the observation that poles (75, 76, 78, 79, 86 and 88; group K) from the craton lie to the north of poles (70, 74 and 87) from area A. The difference between the two groups (≈ 15°) cannot really be regarded as hard evidence. None of the area A poles is highly reliable: for example, only pole 74 was derived from appreciable experimental work and yet the authors could not quote its accuracy to better than 15°. Pole 70 was defined only with low AF treatment data (30 mT) while pole 87, derived from widely scattered directions, is quoted by the authors as a 'tentative pole.' Although the locations of group K poles are, in general, more accurately defined, their ages are somewhat uncertain. Since the poles lie near the Carboniferous-Permian apparent polar path, the possibility that they could be of that age cannot be entirely discounted. However this argument is rather weak being again based on the practice of dating poles according to their locations. The fact is that only pole 88 was shown to be derived from a remanence acquired prior to folding which is probably of pre-Permian age [*Van der Voo*, 1979], thereby delimiting its age to the Devonian-Carboniferous. However, a recent pole (90) derived from a remanence acquired after folding (possibly of Permian age) also falls into group K, entangling the age determination of that group. This begs the question: should those poles be regarded as a group? Not all poles were derived from the same amount of experimental work. In particular, poles 86 and 88 were obtained after considerable experimental work which provides one with reasonable evidence for an initial remanence. Supporting evidence is provided by the fact that both normal and reverse polarity remanences were identified in both studies indicating

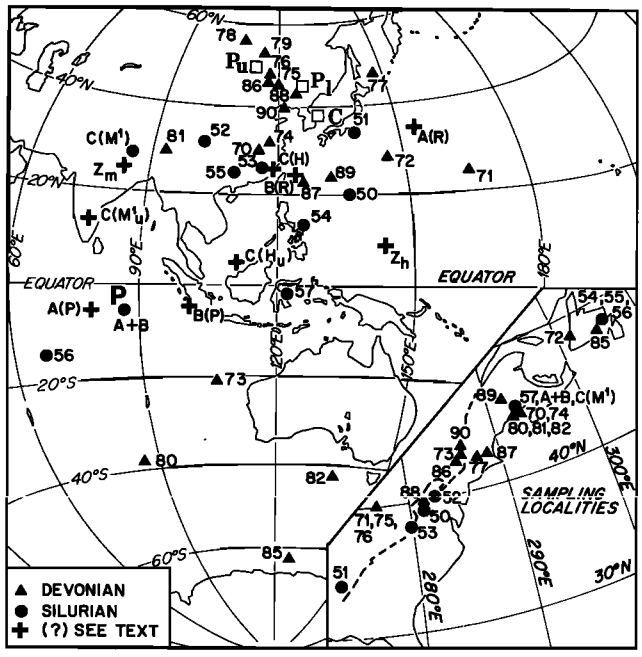


Fig. 14. Poles derived from directions of remanence observed in Devonian and Silurian rock units. It is stressed that the pole age does not necessarily correspond to the rock age. The numbered poles are keyed to the listing of *Roy et al.* [1979]. Additional poles are 53 [*French and Van der Voo,* 1977], 87 [*Schutts et al.,* 1976], 88 [*Van der Voo et al.,* 1979], 89 [*Brown,* 1979], 90 [*Kent,* 1979] and 70 [*Black,* 1964, Perry Formation mean]. For convenience, references to other poles discussed here are 52 [*Roy et al.,* 1967], 54, 55 and 56 [*Lapointe,* 1979], 57, 80 and 81 [*Roy et al.,* 1979], 71, 75 and 76 [*Martin,* 1975], 74 [*Robertson et al.,* 1968], 77 [*Ashwall and Hargraves,* 1977], 78 and 79 [*Elston and Bressler,* 1977] and 86 [*Kent and Opdyke,* 1978]. Open squares: *C* is Carboniferous mean [*Roy et al.,* 1979] and P_l and P_u are Lower and Upper Permian means, respectively [*Van der Voo and French,* 1974]. The reliability and/or age of poles with the plus symbols (this paper) are discussed in the text; the coding being as follows: A, B and C refer to that particular group, (P) to the *P* direction, (R) to the *R* direction, (*M'*) and (*H*) with and without the lowerscript (*u*) to those remanences with or without correction for tilt respectively, Z_m and Z_h to the (assumed) magnetite and hematite remanences of the intrusion. In the inset, the sampling localities from where the poles are obtained are shown except for localities 248°E, 35°N (pole 78) and 248°E, 36°N (pole 79). The dashed line is the approximate western limit of the Appalachian orogen.

that some of the remanence was acquired prior to the magnetic horizon marker (from dual polarity to single (reverse) polarity mode) which for North America has been pinpointed to the early Pennsylvanian (or near the Westphalian-Namurian boundary) [*Roy,* 1977]. It is, therefore, quite possible that the agreement among poles of group K is fortuitous, some of them being of Devonian age and others of Carboniferous, Permian or some other age. If so, this would shatter the correlations attempted on the basis that they form a group [e.g., *Kent and Opdyke,* 1978]. (Lower Carboniferous results have also been used to support detachment of area A, but there again the hypothesis suffers from indetermination of the remanence acquisition age. This problem is discussed in detail by *Roy and Morris* [1981].)

Perhaps the best evidence of area A being much further to the south is to be found in the poles which plot in southern latitudes, for they could be recording motions (of fragments?) of North America resulting from interactions of major plates [*Wilson,* 1966; *Irving,* 1979]. For example, it is possible that pole 80 indicates that southwestern New Brunswick was indeed further to the south near the Siluro-Devonian boundary with pole 81 recording its accretion to the craton. One cannot help but wonder if the longitudinal stringing of Devonian poles along (and near) the 30°N latitude is not the signature of a continental strike slip collision capable of producing local rotations or tilting which could perhaps also account for other poles like 77 as suggested by *Ashwall and Hargraves* [1977].

Furthermore, the extent and integrity of area A are as yet unknown as illustrated by a recent result of *Brown* [1979, pole 89] which requires an extension of the area to include the Appalachians of northern Maine. Even the relationship between 'stable' North American and area A which includes at least part of the Appalachians is unknown. Very few poles have been obtained from west of the Appalachians (Figure 14, inset) and difficulties in interpretation are already apparent. For example, two (75 and 76) of the poles belonging to group K, which is questionably assumed to represent the Devonian location of cratonic North America, were obtained from a stratigraphic column which also yields pole 71 located 42° away.

It is evident from this discussion that numerous speculations (adopting different interpretations for some poles, e.g., group K) can be made owing to the limited data base. We shall indulge in one more relating directly to the results of this investigation. It could be argued that the apparent polar path executed a clockwise movement during the Upper Silurian as described by poles *C*(*M'*), 52, 55, 53, 54, 57, A + B and 56 (leaving out the earlier poles (50 and 51) obtained from less experimental work even though they fit the pattern). It could even be put forward that the direction of the path is in agreement with both poles sequences: B → A (group B being older than group A)—although the results cannot be differentiated statistically—and the poorly substantiated *C*(*M'*) → *C*(*H*). Since the poles forming this path were obtained from central Appalachians, the eastern seaboard and Newfoundland, it could be inferred that the configuration of the area in Silurian time was identical to the present one. However before drawing a 'tentative' polar path to emphasize such far reaching conclusions, it appears to us preferable to await the presentation of more data.

SYNOPSIS

Of the poles derived from this investigation, only poles (*A* + *B*) (*P*) (087°E, 05°S) and *C*(*M'*) (085°E, 28°N) with probable Pembroke and Waweig ages, respectively, are regarded as reliable for apparent polar path construction. All the other poles listed in the tables and shown by plus symbols in Figure 14 are given for completeness: they are classified as false (*B*(*R*), *B*(*R'*), *A*(*R*) and *A*(*R'*)), of indeterminate age or accuracy (*C*(*H*), Z_m and Z_h) or unlikely alternatives (*C*(M_u') and *C*(H_u)).

The investigation reveals that the rocks have recorded much valuable information, only part of which could be recovered by a particular technique. Whether the presently available techniques were sufficient to uncover all the information is, of course, unknown. However, it is clear that the fewer the techniques used, the less the information recovered, and therefore the greater the likelihood of misrepresentation in remanence acquisition age. This could only lead to further ambiguities, or even contribute to a reinforcement syndrome, e.g., the *R* direction being used to support a hypothesis that the apparent pole remained near latitude 30°N during the Silurian.

It is concluded that the tectonic history of the Appalachians and the eastern seaboard can be unraveled by paleomagnetic means. However to do so, it will be necessary to compare data of several detailed investigations where it can be reasonably demonstrated that all the information contained in the rock unit (which hopefully still carries the remanence acquired at the time of formation) has been extracted. Only then will it be possible to reduce the number of options presently available and perhaps associate certain remanences (and poles) with specific events.

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